

Flexible Printed Electronics The Future of Electronics

Electronics is entering a new phase driven by the rapid rise of IoT, wearable technology, smart electronics, and the shift toward miniaturization of products that cannot be served by traditional electronics. A key enabler for this transition is flexible and conformal electronics.

Flexible Printed Electronics is a technology that aims to combine the best of printing and conventional electronics, allowing the building of flexible, soft, and even stretchable electronic products.

Flexible printed electronics allow the development of flexible components such as Flexible Piezoresistive sensors and ultra-thin & flexible Electroluminescent (EL)/LED displays, which remove **form factor constraints and give designers creative freedom.**

According to Salil Musale, Managing Director, Classic Stripes, "For brands and designers to keep innovating and developing next-generation products, they need integrable products that will enable them to develop products that are thin, flexible & exciting to interact with!"

Classic Stripes (a part of Astarc Group), headquartered in Mumbai, is working with companies to provide integrated surface augmentation solutions with thin & flexible Sensing, Illumination, and Touch interface features.

Classic Stripes is one of the largest manufacturers of decorative products in the world. Leveraging its advanced screen printing expertise developed over 3 decades, Classic Stripes has developed multiple Flexible Printed Electronics products. The company is fully equipped to provide end-to-end solutions from design, development, prototyping, testing, to mass production of various products.

The properties of flexible electronics technology such as low weight, mechanical flexibility & durability, simple device integration, along with affordability, make them suitable for a range of applications in various industries, including consumer appliances & durables, modular switches, automobile, consumer electronics, gaming devices, medical devices, wearables & IoT, etc.

EXAMPLES OF APPLICATIONS IN VARIOUS INDUSTRIES

CONSUMER APPLIANCES & DURABLES

A modern coffee machine can utilize touchscreens (transparent touch sensor) as their main human-machine interface (HMI), with the number of physical buttons and switches on the units being kept to a minimum. Optical sensors and force sensors would detect the presence and size of cups in which to dispense the machine's contents.

Sensors on refrigerators are being used in much more practical ways. They can be equipped with force sensors or gesture sensors that can open the door with a gentle touch or wave of your foot, respectively.

MODULAR ELECTRICAL SWITCHES & CONTROLS

The instrument panel with touch buttons having position /pressure sensor and Electroluminescence (EL)/LED backlighting can replace the traditional modular switches and controls. It not only simplifies production by reducing both the number of parts required & the manufacturing steps but also opens up possibilities for design innovation because of the reduced width / thickness of the complete unit.

CONSUMER ELECTRONICS

A typical application of integrated position, force & resistive touch sensing in Consumer Electronics is a multi-touch keypad (mousing solution for OEM and rugged applications). The module's tough moisture-resistant surface can be used with a finger, stylus, or glove; even in harsh environments.



Coffee Machine with Touch-control Panel



Touch-Panel on Refrigerators



Touch-controlled Switches



Automotive Interactive Touch Console



Multi-Touch Keypad



Smart Surface with Touch-activated Backlighting